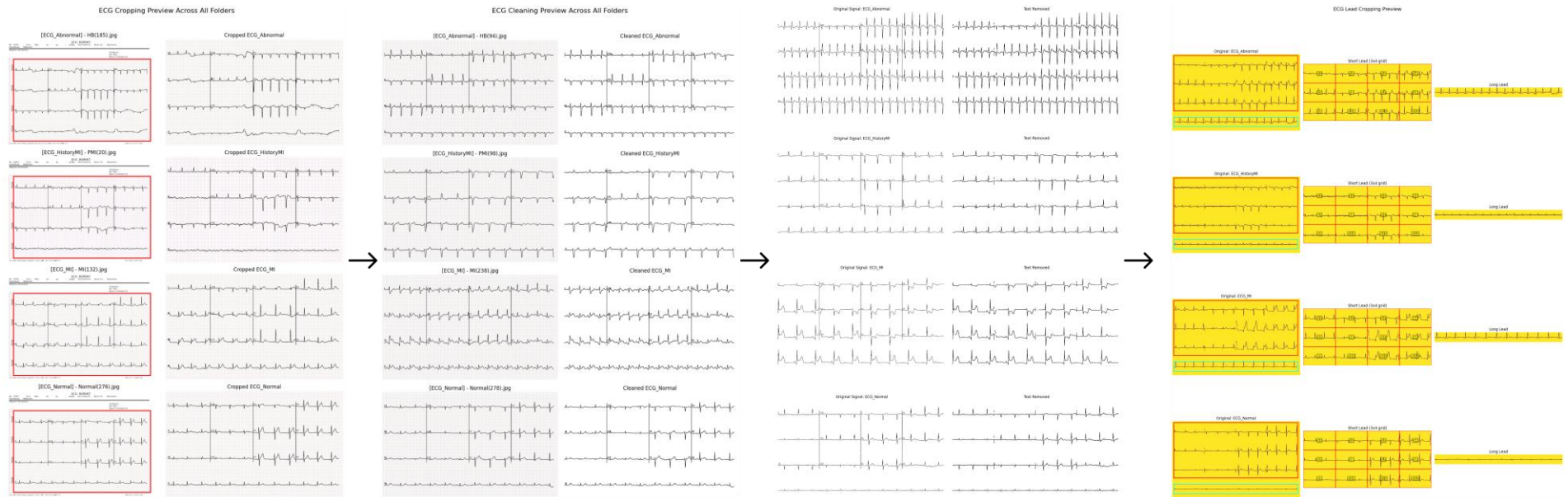


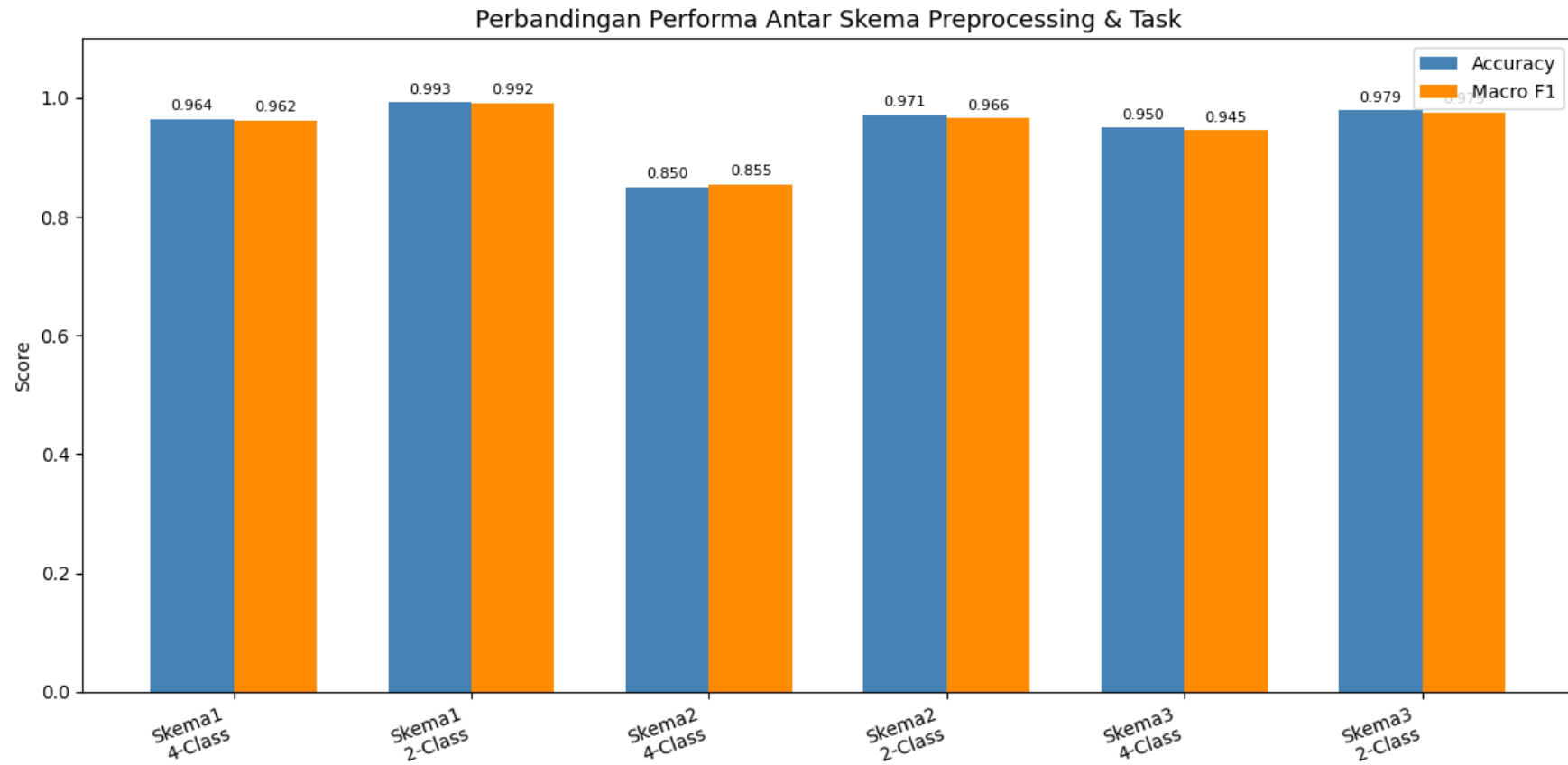
Preprocessing

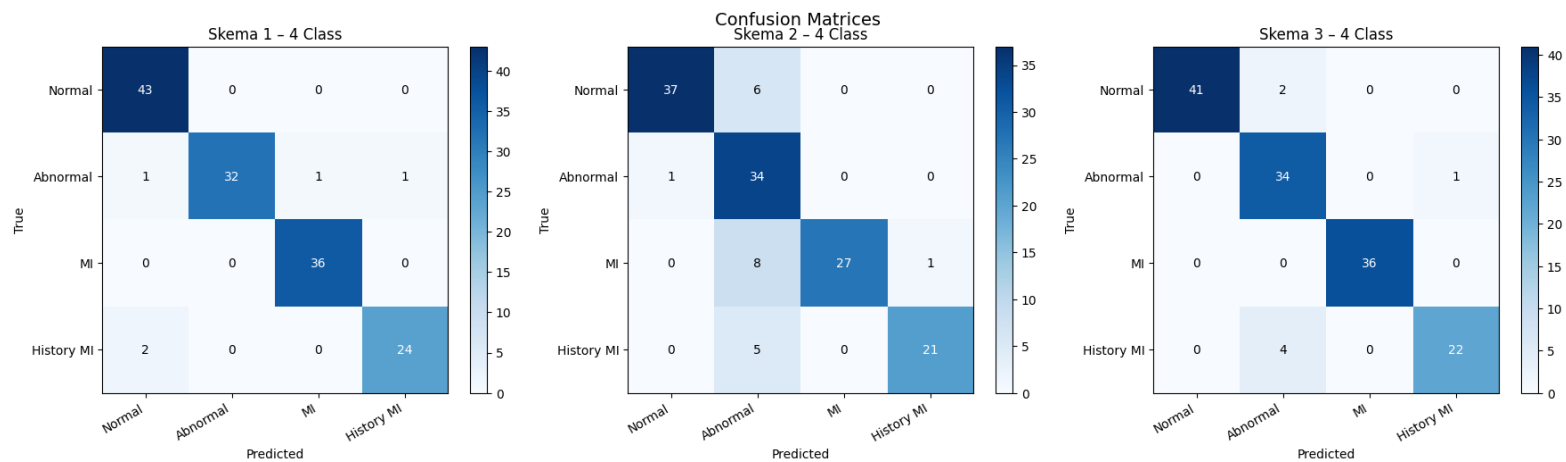
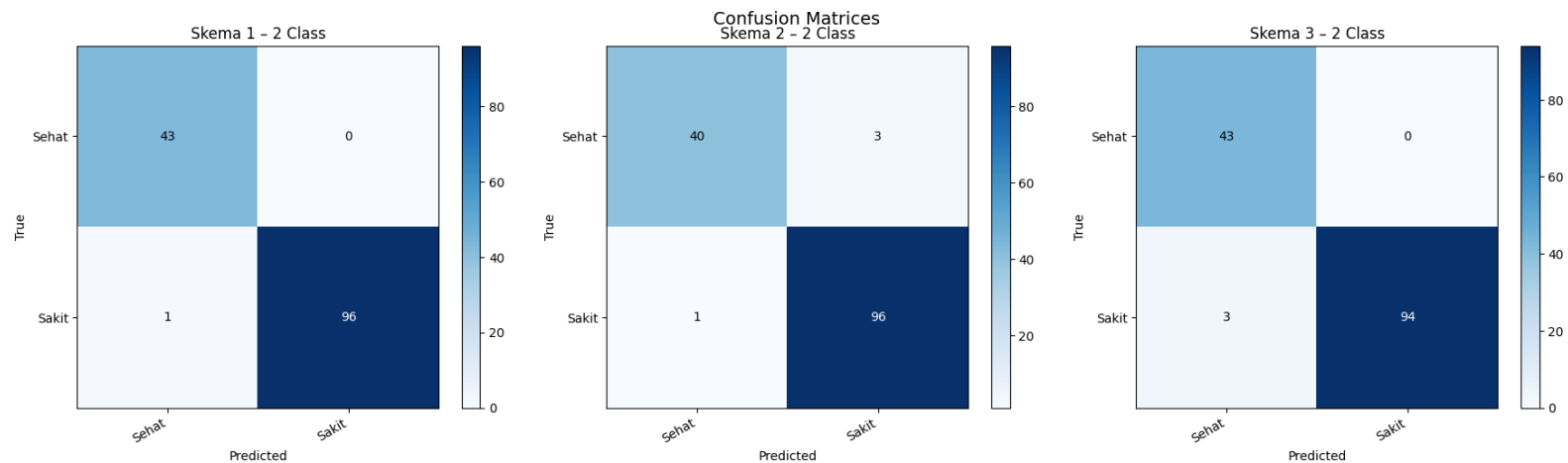


Tahapan preprocessing:

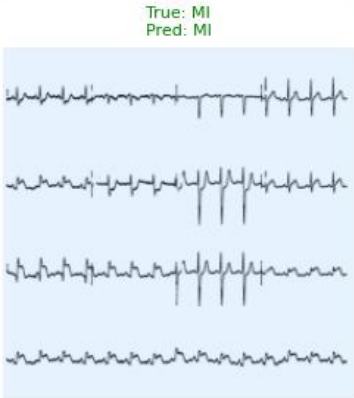
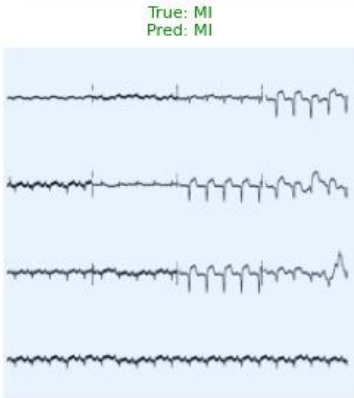
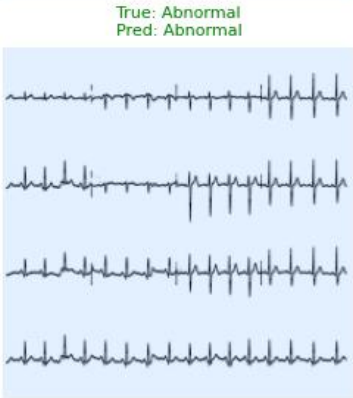
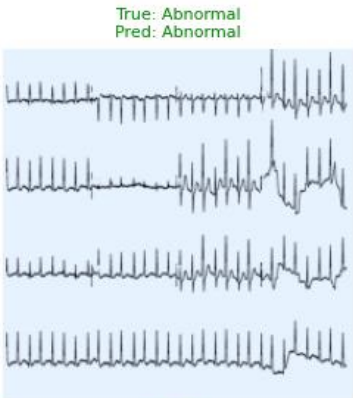
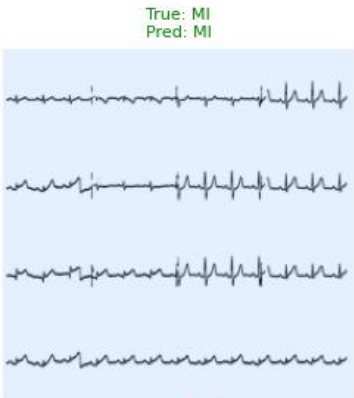
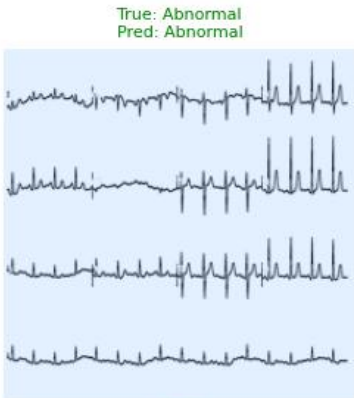
1. Crop area ECG
2. Menghapus grid (background)
3. Menghapus teks dan garis pembatas short lead
4. Cropping menjadi short dan long lead

Modeling Menggunakan ResNet-18





Sample Predictions



Penjelasan Skema:

Dimensi Eksperimen

Dimensi	Nilai	Total Kombinasi
Skema Preprocessing	3 (Scheme 1, 2, 3)	6 kombinasi
Task Klasifikasi	2 (4-class, 2-class)	

Tabel Ringkas

Skema	Input Representation	In-Channels	Sumber Data
Scheme 1	1 gambar clean ECG (full image)	3	data/preprocessed/clean_ecg_signal/
Scheme 2	12 short leads + 1 long lead (13 total)	39 (13×3)	data/preprocessed/cropped_leads/
Scheme 3	6 limb leads + 6 precordial leads + 1 long lead (13 total)	39 (13×3)	data/preprocessed/cropped_leads/

Task Klasifikasi

1. 4-Class (Multi-class):

- Normal (0)
- Abnormal (1)
- MI – Myocardial Infarction (2)
- History MI (3)

2. 2-Class (Binary):

- Sehat (Healthy) (0) – includes Normal
- Sakit (Diseased) (1) – includes Abnormal, MI, History MI